SRI CHAITANYA EDUCATIONAL INSTITUTIONS

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

NEET GRAND TEST-6

Name :..... Hall Ticket No: | | | | | | | |

INSTRUCTIONS TO CANDIDATES

1. The Model **NEET- 2020** is of **3 Hrs** duration. **Time: 02.00 PM – 05.00 PM**.
2. The question paper for NEET-2020 consists of 180 questions comprising 45 questions in Botany, 45 in Zoology, 45 in Physics and 45 in Chemistry for NEET.
3. All questions are of objective type (Multiple choices only)
4. Each question carries four marks.
5. Negative marking: one mark will be deducted for every wrongly answered question.
6. Total Marks 720.
7. The candidates are prohibited from carrying any paper to the examination hall except **HALL TICKET**.
8. No Calculators, Mini-Cards, Watches with Calculators, Pager, Cell Phone, Slide rules or outer aids to calculation will be allowed in the examination hall.
9. Candidates once admitted will not be allowed to leave the hall till half an hour before the closing of the test.
10. A separate sheet is attached in the middle of this booklet for rough work, you can detach and use it.
11. A detachable answer sheet with 180 question blocks, with 4 circles corresponding to 4 multiple choice for each question will be provided. Use HB Pencil to darken the appropriate circle against the question number provided in the sheet. Answer should be marked only on the answer sheet, but not on the question paper booklet.

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

Sec: SR MEDICAL ALL & LTC-IC ***Date : 27.04.20***

TIME : 3 Hrs. ***NEET GRAND TEST – 06*** ***Max.Marks : 720***

1. Identify the incorrect match

Organism	Feature
(1) *Spirogyra*	– Non flagellated isogametes
(2) *Ectocarpus*	– Filamentous form
(3) *Porphyra*	– Unicellular algae used to grow microbes
(4) *Chlamydomonas*	– Microscopic unicellular form

2. Type of life cycle in mosses is
   (1) Haplodiplontic
   (2) Haptontic
   (3) Diplontic
   (4) Haplo diplobiontic
3. What is common between China rose and Tomato?
   (1) Presence of twisted aestivation in corolla
   (2) Presence of axile placentation
   (3) Presence of epigynous flowers
   (4) Presence of monoadelphous stamens
4. Arrange the following structures of a typical chordate in correct sequence from dorsal to ventral side
   (A) Notochord
   (B) Heart
   (C) Nerve cord
   (D) Gut
   (1) A C D B
   (2) C A B D
   (3) C A D B
   (4) A C B D
5. Identify the correct one related to *Pheretima*

(1)	Typhlosole	Mid ventral fold of intestinal wall to increase area of absorption
(2)	Intestinal caecae	Pair of conical projections at the junction of stomach and intestine
(3)	Blood glands	Found in 4, 5, 6 segments and produce RBC and phagocytic cells
(4)	Clitellum	14-16 segments in which forests of integumentary nephridia are found

6. During breathing, diaphragm is arched upwards due to
   (1) contraction of phrenic muscles
   (2) contraction of external intercostals
   (3) relaxation of diaphragm
   (4) upward movement of ribs and sternum
7. Mark the correctly matched pair.
   (1) $C_{1+2+(2)}$ – Mulathi
   (2) $A_{(9)+1}$ – Gloriosa
   (3) P_{3+3} – Belladonna
   (4) $G_{(2)}$ – Indigofera

8. Protonephridia are the excretory structures in
   (1) Lancelets, Rotifers
   (2) *Fasciola, Gorgonia*
   (3) *Planaria, Doliolum*
   (4) *Amphioxus, Antedon*

9. The prokaryotic cell structure exemplified by both of the statements A and B is
   A. Not bounded by any membrane and lie free in the cytoplasm.
   B. Stores reserve material in the cytoplasm.
   (1) Ribosomes
   (2) Mesosome
   (3) Chromatophores
   (4) Inclusion bodies

10. Identify the steroid hormones that stimulate the RBC production
   (a) Erythropoietin
   (b) Cortisol
   (c) Thyroxine
   (d) Androgens
   (1) a b c
   (2) b c d
   (3) b d
   (4) a b c d

11. Deficiency of which group of elements cause inhibition of cell division?
   (1) N, K, S, Mo
   (2) N, S, Mo, Cu
   (3) Ca, Mg, Cu, K
   (4) Fe, Mn, Mo, B

12. Which of the following disorders can not be inherited by a man from his father?
   (1) Myotonic dystrophy
   (2) Hypertrichosis
   (3) Huntington's chorea
   (4) Haemophilia

13. In the Z-scheme of photosynthesis
   (1) The two photosystems work in series, first PS-I and then PS-II
   (2) The electrons move from H_2O to $NADP^+$
   (3) Protons accumulate in stroma is called proton reservoir
   (4) Oxygen is released on the stroma side of thylakoid membrane

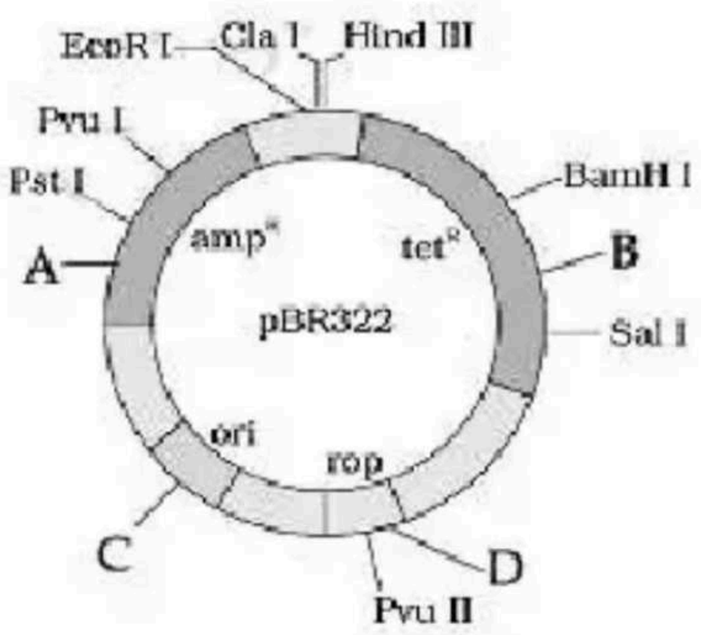
14. How many ATP are produced in mitochondria by substrate level phosphorylation only if two molecules of pyruvate get oxidised completely?
   (1) Two
   (2) Four
   (3) One
   (4) Six

15. Appearance of dry, scaly lesions on the skin with intense itching is the main symptom of the disease caused by
   (1) *Babesia*
   (2) *Microsporum*
   (3) *Streptococcus*
   (4) *Mycoplasma*

16. Very small animals are rarely found in polar regions, because
    (1) small animals have large volume relative to their surface area
    (2) In small animals heat generation far exceeds the heat loss
    (3) small animals have larger surface area relative to their volume
    (4) all the small animals are thermo conformers
17. Which of the following is not an outbreeding device?
    (1) Presence of dichogamy
    (2) Synchronisation in pollen release and stigma receptivity
    (3) Presence of self incompatibility
    (4) Presence of unisexual flowers
18. Biomass available for the consumption to herbivores and decomposers is
    (1) Gross primary productivity
    (2) Net primary productivity
    (3) Gross secondary productivity
    (4) Net secondary productivity
19. Which of the following genotype is a dihybrid?
    (1) AAbb
    (2) PPQqrr
    (3) TtRrYY
    (4) EeFfGgHH
20. Radula functions as the rasping organ in
    (1) Sea mouse
    (2) Sea horse
    (3) Sea cow
    (4) Sea hare
21. Which of the following proteins will be produced in a *lac* operon, when lactose is available in the medium of the bacteria?
    (i) β -galactosidase
    (ii) Transacetylase
    (iii) Permease
    (1) Only (i)
    (2) Only (ii) & (iii)
    (3) Only (i) & (iii)
    (4) All of these
22. In cockroach, ootheca is formed by the secretion of
    (1) Utricular gland
    (2) Conglobate gland
    (3) Cowper's gland
    (4) Collaterial gland
23. In wheat, resistance to cereal leaf beetle is due to
    (1) Biochemical adaptation
    (2) Physiological adaption
    (3) Behavioural adaptation
    (4) Morphological adaptation
24. In humans, pCO_2 of systemic arteries is
    (1) more than that of pulmonary arteries
    (2) more than that of pulmonary veins
    (3) same as that of pulmonary arteries
    (4) same as that of alveoli

25. Isolated protoplasts from two different plant cells can be fused to obtain
    (1) Monoclonal cultures
    (2) Somatic hybrids
    (3) Purelines
    (4) Hybrid vigour

26. Given below is a diagrammatic representation of *E. coli* cloning vector pBR-322.



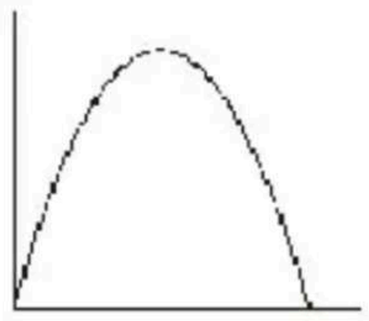
Which of the following labelled structure is responsible for controlling the copy number of linked DNA?
    (1) A
    (2) B
    (3) C
    (4) D

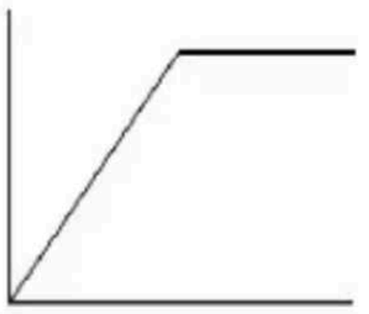
27. Choose the incorrect match related to human kidney.

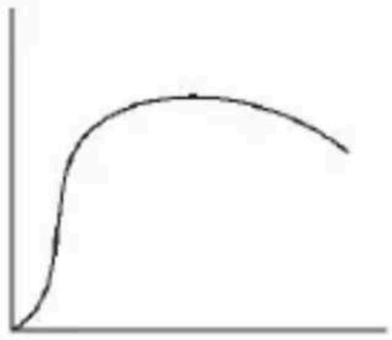
(1)	Conditional reabsorption of sodium and water	Distal convoluted tubule
(2)	Reabsorption of urea into the medullary interstitium	Ascending limb of loop of Henle
(3)	Selective secretion of K^+ and H^+ ions	Collecting duct
(4)	Reabsorption of glucose and amino acids	Proximal convoluted tubule

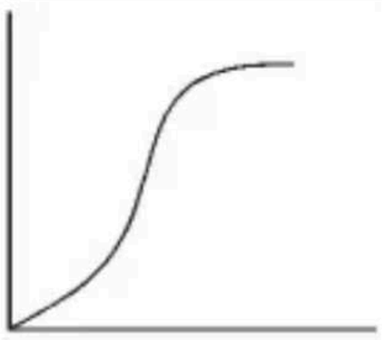
28. Sympathetic neural system stimulates
    (1) Secretion of digestive enzymes
    (2) Contraction of radial muscles of iris
    (3) Constriction of bronchi
    (4) Contraction of circular muscles of iris

29. Which one of the following explains the effect of temperature on enzyme action?

(1) 

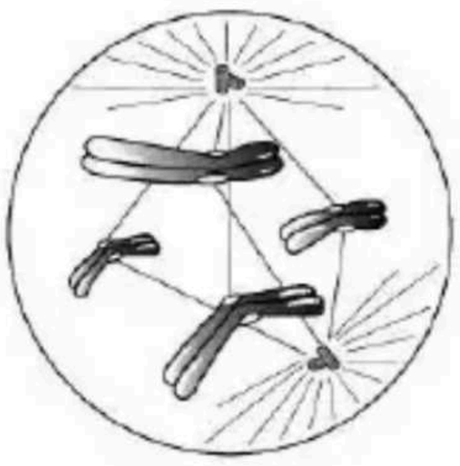
(2) 

(3) 

(4) 

30. In human male reproductive system, ejaculatory duct is formed by
    (1) Vas deferens and ureter
    (2) Vas deferens and duct of prostate gland
    (3) Vas deferens and duct of urinary bladder
    (4) Vas deferens and duct of seminal vesicle

31. Find set of fungi belonging to ascomycetes
    (1) Morels, *Albugo*, *Agaricus*
    (2) *Neurospora*, *Agaricus*, Morels
    (3) *Aspergillus*, *Colletotrichum*, *Penicillium*
    (4) *Claviceps*, *Neurospora*, *Aspergillus*
32. Identify the protozoan that causes a sexually transmitted disease
    (1) *Chlamydia*
    (2) *Trichophyton*
    (3) *Trichomonas*
    (4) *Treponema*
33. Some structures are given below Protonema cell, Prothallus cell of fern, PEN of angiosperm, Gemma of Marchantia. Select correct sequence of their respective ploidy level
    (1) n, n, 3n, n
    (2) 2n, 2n, 3n, 2n
    (3) n, n, n, n
    (4) 2n, 2n, n, 2n
34. The figure shown below depicts which stage of cell division



    (1) Metaphase
    (2) Prophase
    (3) Transition to metaphase
    (4) Anaphase
35. The earliest organisms that appeared on the earth were
    (1) anaerobic chemoheterotrophs
    (2) aerobic chemoautotrophs
    (3) anaerobic chemoautotrophs
    (4) anaerobic photoautotrophs
36. The practice of mating between Leghorn and Australorp is
    (1) Inbreeding
    (2) Out crossing
    (3) Interspecific hybridisation
    (4) Cross breeding
37. Find odd one out w.r.t. ethylene functions
    (1) Rapid internode elongation in deep water rice plants
    (2) Initiates flowering and synchronising fruit
    (3) Thinning of fruits like cherry, walnut and cotton
    (4) Apical growth and adventitious shoot formation
38. Ecological pyramid is inverted in the case of
    (1) Pyramid of numbers in grassland
    (2) Pyramid of biomass in sea
    (3) Pyramid of energy in forest
    (4) Pyramid of numbers in a pond
39. Majority of insect pollinated flowers are
    (1) Nectarless, odourless and coloured
    (2) Colourful, fragrant and nectar rich
    (3) Coloured, fragrant and nectarless
    (4) Coated by a waxy layer

40. Identify the correct representation of number of species of different taxa
    (1) Fungi > Algae > Lichens > Mosses
    (2) Fungi > Fishes > Birds > Reptiles
    (3) Fishes > Mammals > Birds >Amphibians
    (4) Angiosperms > Ferns > Algae > Lichens
41. Which of the following structures present in both DNA and RNA
    a) Ribose
    b) Thiamine
    c) Sugar phosphate chain
    d) Double helix structure
    e) Adenine
    (1) a & c
    (2) c & b
    (3) c & e
    (4) b & c
42. Living mechanical tissue is mostly restricted to the _____ parts of _____ plant organs in dicots
    (1) Epidermal, sub-aerial
    (2) Sub-epidermal, aerial
    (3) Epidermal, aerial
    (4) Sub-epidermal, sub-aerial
43. Body is covered by dry and cornified skin with epidermal scales in
    (1) *Hemidactylus, Hippocampus*
    (2) *Calotes, Chameleon*
    (3) *Corvus, Clarias*
    (4) *Psittacula, Pterophyllum*
44. In the absence of one of the enzymes of succus entericus, protein digestion would be severely impaired. Identify that enzyme
    (1) trypsin
    (2) chymotrypsin
    (3) enterokinase
    (4) all the above
45. Growth and differentiation in plants is _____ and ____ respectively.
    (1) Closed, open
    (2) Closed, closed
    (3) Open, closed
    (4) Open, open
46. In a cardiac cycle, closure of semilunar valves takes place during
    (1) Atrial systole
    (2) Ventricular systole
    (3) Joint systole
    (4) Ventricular diastole
47. How many of the organisms show external fertilization?
    (1) Bryophytes & Pteridophytes
    (2) Fungi & Gymnosperms
    (3) Higher animals
    (4) Aquatic alga, fishes & amphibians
48. Read the following statements related to the skeletal muscle and choose the correct option.
    (I) Binding of calcium with a sub unit of troponin, removes the masking of actin binding sites on myosin
    (II) After muscle contraction, Ca^{++} ions are pumped back to the sarcoplasmic cisternae which results in masking of active sites of actin filaments
    (1) Both I, II are correct
    (2) Only I is correct
    (3) Both I, II are incorrect
    (4) Only II is correct

49. Genes for which of the following characters in *Drosophila* were found to be very tightly linked by Morgan?
(1) Eye colour and wing size
(2) Body colour and wing size
(3) Wing size and eye colour
(4) Body and eye colour

50. In the umbilical cord of human embryo, blood vessels
(1) are absent
(2) contain foetal blood
(3) contain maternal blood
(4) both 2 and 3

51. Microbes are useful to man in diverse ways. Which of the following products are formed by fermentive activity of yeast?
(1) Bread, curd, cheese
(2) Toddy, cheese, curd
(3) Biogas, biofertilizer, cheese
(4) Bread, ethanol, wine

52. Which of the following syndromes is an example for allosomal trisomy?
(1) Down's syndrome
(2) Turner's syndrome
(3) Edward's syndrome
(4) Klinefelter's syndrome

53. The pioneers and stable climax community formed in hydrosere are respectively
(1) Lichens, trees
(2) Mosses, grasses
(3) Sedges, grasses
(4) Phytoplanktons, trees

54. Theory proposed by Lamarck was
(1) continuity of germplasm
(2) pangenesis
(3) survival of the fittest
(4) inheritance of acquired characters

55. Viroids are
(1) Larger than viruses
(2) Causal agent of witches broom disease
(3) Having only RNA of low molecular weight
(4) The entity lacking the genetic material

56. Select the incorrect statement w.r.t. wood
(1) Spring wood has large number of xylary elements
(2) Late wood has broad vessels
(3) Autumn wood is darker and has higher density
(4) Early wood is formed due to high activity of cambium.

57. Which of the following is an example for allergy?
(1) Scarlet fever
(2) Typhus fever
(3) Hay fever
(4) Trench fever

58. The best evidence in support of Gause's competitive exclusion principle is the interaction between
(1) *Chathamalus and Balanus*
(2) Abingdon tortoise and goats
(3) Flamingoes and fishes
(4) Warbler birds living on the same tree

59. If a meiocyte contains 20 chromosomes, then what will be the chromosome number at each pole during anaphase-I?
(1) 40
(2) 20
(3) 10
(4) 5

60. What was the cause for the extinction of Steller's sea cow?
(1) Habitat fragmentaion
(2) Co-extinction
(3) Alien species invasion
(4) Over-exploitation

61. Short distance movement of substances does not involve
(1) Cytoplasmic streaming
(2) Energy
(3) Diffusion
(4) Vascular system

62. Emission of poisonous gases from automobiles can be effectively reduced by using
(1) Scrubbers
(2) Electrostatic precipitators
(3) Incinerators
(4) Catalytic converters

63. Which of the following activity is not related directly with stress hormone?
(1) Bud dormancy
(2) Seed dormancy
(3) Cell division
(4) Inhibition of cambial activity

64. Which of the following is incorrect about areolar tissue?
(1) serves as a support frame work for epithelium
(2) contains macrophages, mast cells and fibre secreting cells
(3) it is found beneath the skin
(4) fibres and fibroblasts are compactly packed in the ground substance

65. Which of the following enzymes is not used for extraction of desired gene (pure) from a plant?
(1) Ribonuclease
(2) Protease
(3) Lysozyme
(4) Cellulase

66. Which of the following digestive juices do not contain amylase enzyme?
(a) Pancreatic juice
(b) Bile juice
(c) Gastric juice
(d) Saliva
(e) Intestinal juice
(1) b c d
(2) b c e
(3) a c e
(4) a b c

67. Which of the following is not a direct method of gene transfer?
(1) Microinjection
(2) Biolistic
(3) Electroporation
(4) Cosmid

68. Where does exchange of substances between blood and tissues take place?
(1) arteries
(2) veins
(3) capillaries
(4) all the above

69. RNA Interference technology takes place
(1) In all prokaryotic organisms as a method of cellular defense
(2) During tissue culture
(3) In all eukaryotic organisms as a method of cellular defense
(4) In all eukaryotes as a method of cellular proliferation

70. Which bone articulates with sternum and acromion process?
(1) rib
(2) scapula
(3) clavicle
(4) coracoids

71. A taxon is
(1) A group of related families
(2) A group of related species
(3) A type of living organisms
(4) A taxonomic group of any ranking

72. Peat moss is another name of
(1) *Sphagnum*
(2) *Marchantia*
(3) *Riccia*
(4) *Dryopteris*

73. Match the following

Column-I	Column-II
A) Air (prevention and control of pollution) Act	1) 1986
B) Water (prevention and control of pollution) Act	2) 1981
C) Environment (protection) Act	3) 2002
D) The earth summit on biological diversity	4) 1974
	5) 1992

	A	B	C	D
(1)	1	2	3	4
(2)	2	4	1	5
(3)	1	5	4	3
(4)	3	2	5	1

74. Lens in the eye is held in place by ligaments attached to
(1) Cornea
(2) Iris
(3) Sclera
(4) Ciliary body

75. The cork cambium, cork and secondary cortex are collectively called
(1) Phellderm
(2) Phellogen
(3) Periderm
(4) Phellem

76. Ribozyme is a
(1) Simple protein
(2) RNA
(3) Ribonucleoprotein
(4) Repetitive DNA

77. Choose the correct match.

(1)	Thyrocalcitonin	Steroid hormone	Decreases blood calcium levels
(2)	Melanotropin	Pineal hormone	Plays a role in regulation of 24hr rhythm of the body
(3)	Thymosin	Peptide hormone	Plays important role in differentiation of antibody producing lymphocytes
(4)	Cholecystokinin	Peptide hormone	Stimulates the secretion of pancreatic enzymes

78. Sodium-potassium pumps maintain the ionic gradients across the resting axonal membrane by
(1) passive transport of 3 Na^+ outwards and 2K^+ inwards
(2) active transport of 3 Na^+ outwards and 2K^+ inwards
(3) active transport of 3 Na^+ inwards and 2K^+ outwards
(4) passive transport of 3 Na^+ inwards and 2K^+ outwards

79. The *Triticale* is an intergeneric hybrid between
(1) Wheat and maize
(2) Maize and rye
(3) Wheat and rye
(4) Bajra and wheat

80. In humans, part of the fallopian duct that is closest to the uterus is
(1) ampulla
(2) cervix
(3) infundibulum
(4) isthmus

81. The mode of catching insects in *Drosera* plants is by means of
(1) Sensitive glandular hairs
(2) Sensitive trigger hairs
(3) Leaves which are modified into pitcher
(4) Leaf segments modified into bladder

82. The extinct human who lived 1,00,000 to 40,000 years ago in Europe and Asia was
(1) *Homo erectus*
(2) Neanderthal man
(3) Modern *Homo sapiens*
(4) Australopithecus

83. Photorespiration is favoured by
(1) High O_2 and low CO_2
(2) Low light and high O_2
(3) Low temperature and high O_2
(4) Low O_2 and high CO_2

84. Type of immunity provided by administering tetanus toxoid is
(1) Natural active immunity
(2) Natural passive immunity
(3) Artificial active immunity
(4) Artificial passive immunity

85. Terminal cytochrome of respiratory chain which donates electrons to oxygen is
(1) Cyt b
(2) Cyt c
(3) Cyt a_1
(4) Cyt a_3

86. Which of the following is correct for r –selected species?
(1) large number of progeny with small size
(2) large number of progeny with large size
(3) small number of progeny with large size
(4) small number of progeny with small size

87. The catalytic efficiency of two different enzymes can be compared by the
(1) Fromation of the product
(2) pH optimum value
(3) Km value
(4) Molecular size of the enzyme

88. Which of the following is not characteristic of a stable community

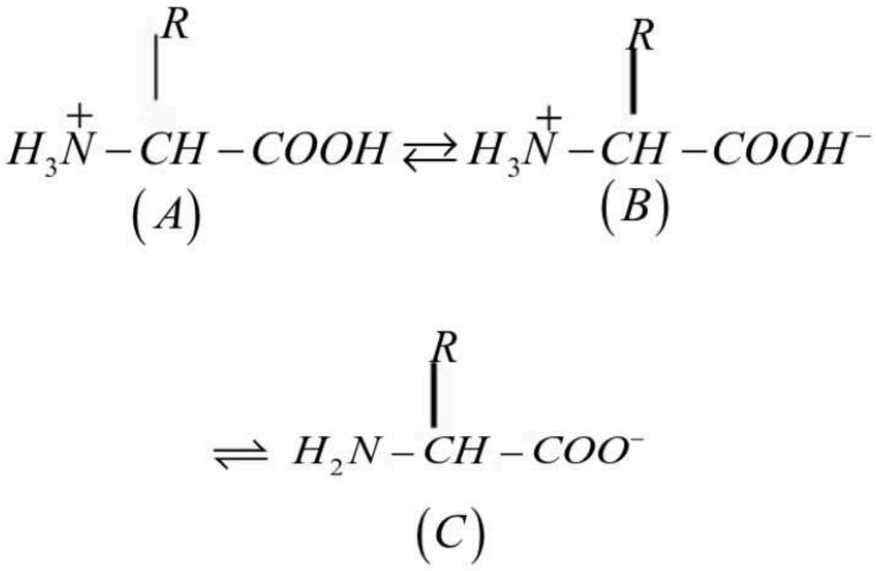
(1) more resilient to occasional disturbances

(2) shows more species diversity

(3) shows too much variation in productivity year to year

(4) more resistant to invasion by alien species

89.



Which of the above is Zwitter ionic form

(1) A

(2) C

(3) B

(4) All of these

90. Choose the correct option about CFCs

(1) contribute to global warming

(2) cause depletion of ozone in the trophosphere

(3) catalyse conversion of O_2 to O_3

(4) both 1 and 2

91. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and 25th division of circular scale coincides with the reference level. If screw gauge has a zero error of -0.004 cm, the correct diameter of the ball is

(1) 0.521 cm

(2) 0.525 cm

(3) 0.053 cm

(4) 0.529 cm

92. The position x of a particle with respect to time t along x-axis is given by $x = 9t^2 - t^3$ where x is in metres and t in seconds. What will be the position of this particle when it achieves maximum speed along the $+x$ direction?

(1) 54 m

(2) 81 m

(3) 24 m

(4) 32 m

93. If vectors $\vec{A} = \cos\omega t\hat{i} + \sin\omega t\,\hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other is

(1) $t = \frac{\pi}{\omega}$

(2) $t = 0$

(3) $t = \frac{\pi}{4\omega}$

(4) $t = \frac{\pi}{2\omega}$

94. A car is negotiating a cured road of radius R. The road is banked at an angle θ. The coefficient of friction between the tyres of the car and the road is μ_s. The maximum safe velocity on this road is

(1) $\sqrt{\frac{g}{R}\left(\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}\right)}$

(2) $\sqrt{\frac{g}{R^2}\left(\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}\right)}$

(3) $\sqrt{gR^2\left(\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}\right)}$

(4) $\sqrt{gR\left(\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}\right)}$

95. In a rocket, fuel burns at the rate of 1 kg/s. This fuel is ejected from the rocket with a velocity of 60 km/s. This exerts a force on the rocket equal to

(1) 6000 N

(2) 60000 N

(3) 60 N

(4) 600 N

96. The potential energy of a particle in a force field is $U=\frac{A}{r^2}-\frac{B}{r}$ where A and B are positive constants and r is the distance of particle from the centre of the field. For stable equilibrium, the distance of the particle is

(1) $\frac{B}{2A}$

(2) $\frac{2A}{B}$

(3) $\frac{A}{B}$

(4) $\frac{B}{A}$

97. Three objects, A : (a solid sphere), B : (a thin circular disk) and C : (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation

(1) $W_C > W_B > W_A$

(2) $W_A > W_B > W_C$

(3) $W_B > W_A > W_C$

(4) $W_A > W_C > W_B$

98. A circular platform is mounted on a frictionless vertical axle. Its radius R = 2 m and its moment of inertia about the axle is $200 kgm^2$. It is initially at rest. A 50 kg man stands on the edge of the platform and begins to walk along the edge at the speed of $1 ms^{-1}$ relative to the ground. Time taken by the man to complete one revolution is

(1) πs

(2) $\left(\frac{3\pi}{2}\right)s$

(3) $(2\pi)s$

(4) $\left(\frac{\pi}{2}\right)s$

99. If the mass of the sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is not correct ?

(1) Raindrops will fall faster

(2) Walking on the ground would become more difficult

(3) Time period of a simple pendulum on the Earth would decrease

(4) g on the Earth will not change

100. The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_0. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. The value of n is

(1) $\frac{3}{4}$

(2) $\frac{4}{3}$

(3) $\frac{256}{81}$

(4) $\frac{81}{256}$

101. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied?

(1) length = 200 cm, diameter = 2 mm

(2) length = 300 cm, diameter = 3 mm

(3) length= 50 cm, diameter = 0.5 mm

(4) length = 100 cm, diameter = 1 mm

102. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

(1) $15\,RT$

(2) RT

(3) $11\,RT$

(4) $4\,RT$

103. A Carnot engine, having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is

(1) 90 J

(2) 1 J

(3) 100 J

(4) 99 J

104. When 1 kg of ice at 0°C melts to water at 0°C, the resulting change in its entropy, taking latent heat of ice to be 80 cal / g, is

(1) 273 cal/K

(2) $8\times10^4\,cal\,/\,K$

(3) 80 cal/K

(4) 293 cal/K

105. Out of the following functions representing motion of a particle, which represents SHM?

a) $y=\sin\omega t-\cos\omega t$

b) $y=1+\omega t+\omega^2t^2$

(1) Both (a) and (b)

(2) Only (b)

(3) Only (a)

(4) Neither (a) nor (b)

106. A body is executing simple harmonic motion. When the displacements from the mean position is 4 cm and 5 cm, the corresponding velocities of the body are 10 cm/sec and 8 cm/sec. Then the time period of the body is

(1) 2πsec

(2) $\pi/2$sec

(3) πsec

(4) $3\pi/2$sec

107. Two cars moving in opposite directions approach each other with speed of $22ms^{-1}$ and $16.5ms^{-1}$ respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is (velocity of sound is $340ms^{-1}$)

(1) 361 Hz

(2) 411 Hz

(3) 448 Hz

(4) 350 Hz

108. The wave described by $y=0.25\sin\left(10\pi x-2\pi t\right)$, where x and y are in meters and t in seconds, is a wave travelling along the

(1) +ve x direction with frequency 1 Hz and wavelength $\lambda=0.2m$

(2) -ve x direction with frequency 0.25 m and wavelength $\lambda=0.2m$

(3) -ve x direction with frequency 1 Hz

(4) +ve x direction with frequency π Hz and wavelength $\lambda=0.2m$

109. The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

(1) Independent of the distance between the plates

(2) Linearly proportional to the distance between the plates

(3) Proportional to the square root of the distance between the plates

(4) Inversely proportional to the distance between the plates

110. A series combination of n_1 capacitors, each of value C_1, is charged by a source of potential difference 4V. When another parallel combination of n_2 capacitors, each of value C_2, is charged by a source of potential difference V, it has the same (total) energy stored in it, as the first combination has. The value of C_2, in terms of C_1, is then

(1) $\frac{2C_1}{n_1n_2}$

(2) $16\frac{n_2}{n_1}C_1$

(3) $2\frac{n_2}{n_1}C_1$

(4) $\frac{16C_1}{n_1n_2}$

111. A galvanometer with a coil of resistance 12Ω shows a full scale deflection for a current of 2.5 mA. It can be converted into an ammeter of range 7.5A by connecting a resistance of

(1) $4\times10^{-3}\Omega$ in series

(2) 2988Ω in series

(3) $4\times10^{-3}\Omega$ in parallel

(4) 2988Ω in parallel

112. A carbon resistor of $\left(47\pm4.7\right)k\Omega$ is to be marked with rings of different colours for its identification. The colour code sequence will be

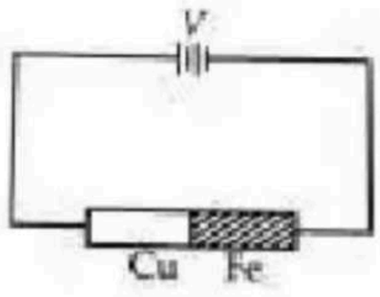
(1) Violet – Yellow – Orange – Silver

(2) Yellow – Violet – Orange – Silver

(3) Yellow – Green – Violet – Gold

(4) Green – Orange – Violet – Gold

113. Two rods are joined end to end, as shown. Both have a cross – sectional area of $0.01 cm^2$. Each is 1 meter long. One rod is of copper with a resistivity of 1.7×10^{-6} ohm-centimeter, the other is of iron with a resistivity of 10^{-5} ohm-centimeter.



How much voltage is required to produce a current of 1 ampere in the rods ?

(1) $0.00145V$

(2) $0.0145V$

(3) $1.7\times10^{-6}V$

(4) $0.117V$

114. A proton and an alpha particle both enter a region of uniform magnetic field B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy of proton is 1 MeV then the kinetic energy of the alpha particle will be

(1) 1.5 MeV

(2) 1MeV

(3) 4 MeV

(4) 0.5 MeV

115. A closely wound solenoid of 2000 turns and area of cross-section $1.5\times10^{-4}m^2$ carries a current of 2.0 A. It is suspended through its centre and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform magnetic field 5×10^{-2} tesla making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be

(1) $3\times10^{-3}Nm$

(2) $1.5\times10^{-3}Nm$

(3) $1.5\times10^{-2}Nm$

(4) $3\times10^{-2}Nm$

116. The magnetic susceptibility is negative for

(1) Ferromagnetic material only

(2) Paramagnetic and ferromagnetic materials

(3) Diamagnetic material only

(4) Paramagnetic material only

117. Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication?

(1) $R=15\Omega, L=1.5H, C=35\mu F$

(2) $R=15\Omega, L=2.5H, C=45\mu F$

(3) $R=15\Omega, L=3.5H, C=30\mu F$

(4) $R=15\Omega, L=1.5H, C=45\mu F$

118. Two coils have a mutual inductance 0.005 H. The current changes in the first coil according to equation $I=I_0\sin\omega t$, where $I_0=10A$ and $\omega=100\pi rad/\sec$. The maximum value of e.m.f. in the second coil is

(1) π

(2) 5π

(3) 2π

(4) 4π

119. The structure of solids is investigated by using
(1) Cosmic rays
(2) X-rays
(3) γ-rays
(4) infra-red radiations

120. The interference pattern is obtained with two coherent light sources of intensity ratio n. In the interference pattern, the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be
(1) $\frac{\sqrt{n}}{n+1}$
(2) $\frac{2\sqrt{n}}{n+1}$
(3) $\frac{\sqrt{n}}{(n+1)^2}$
(4) $\frac{2\sqrt{n}}{(n+1)^2}$

121. In Young's double slit experiment the distance between the slits and the screen is doubled. The separation between the slits is reduced to half. As a result the fringe width
(1) Is halved
(2) Becomes four times
(3) Remains unchanged
(4) Remains unchanged

122. If the refractive index of material of equilateral prism is $\sqrt{3}$, then angle of minimum deviation of the prism is
(1) 60°
(2) 45°
(3) 30°
(4) 75°

123. Electro magnetic wave is travelling along y-direction. If the corresponding $\vec{E}$ at any time is along the x-axis then the direction of $\vec{B}$ at that time is along
(1) +y-axis
(2) +x-axis
(3) +z-axis
(4) -z-axis

124. If the momentum of an electron is changed by P, then the de-Brogile's wavelength associated with it changes by 0.5%. The initial momentum of electron will be
(1) $200P$
(2) $400P$
(3) $\frac{P}{200}$
(4) $100P$

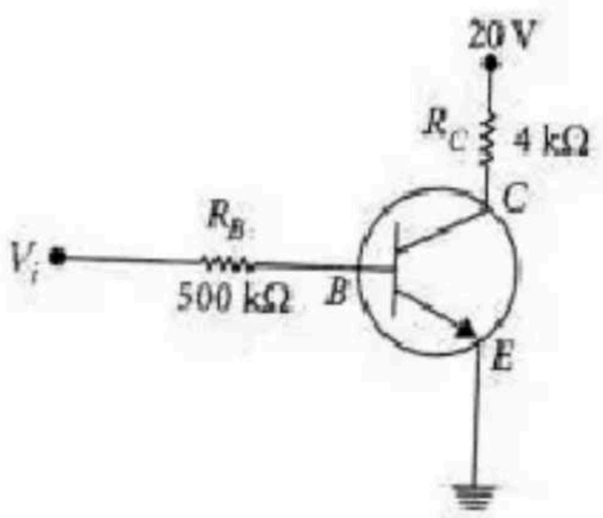
125. The cathode of a photoelectric cell is changed such that the work function changes from W_1 to W_2 $(W_2 > W_1)$. If the current before and after changes are I_1 and I_2, all other conditions remaining unchanged, then (assuming $h\upsilon > W_2$)
(1) $I_1 = I_2$
(2) $I_1 < I_2$
(3) $I_1 > I_2$
(4) $I_1 < I_2 < 2I_1$

126. An electron of a stationary hydrogen atom passes from the fifth energy level to the ground level. The velocity that the atom acquired as a result of photon emission will be (Terms have their usual meaning)
(1) $\frac{24hR}{25m}$
(2) $\frac{25hR}{24m}$
(3) $\frac{25m}{24hR}$
(4) $\frac{24m}{25hR}$

127. Two radioactive substances A and B have decay constants 5λ and λ respectively. At t = 0 they have the same number of nuclei. The ratio of number of nuclei of A to those of B will be $(1/e)^2$ after a time interval

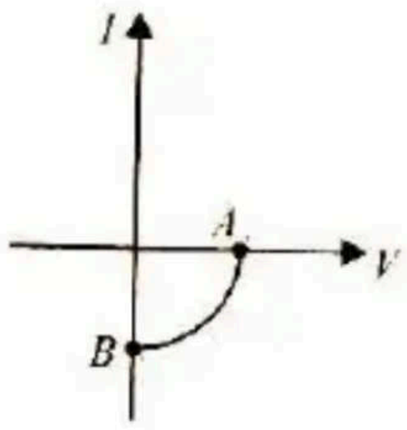
(1) 4λ
(2) 2λ
(3) $1/2\lambda$
(4) $1/4\lambda$

128. In the circuit shown in the figure , the input voltage V_i is 20V, $V_{BE}=0$ and $V_{CE}=0$. The values of I_B, I_C and β are given by



(1) $I_B=40\mu A, I_C=10mA, \beta=250$
(2) $I_B=25\mu A, I_C=5mA, \beta=200$
(3) $I_B=20\mu A, I_C=5mA, \beta=250$
(4) $I_B=40\mu A, I_C=5mA, \beta=125$

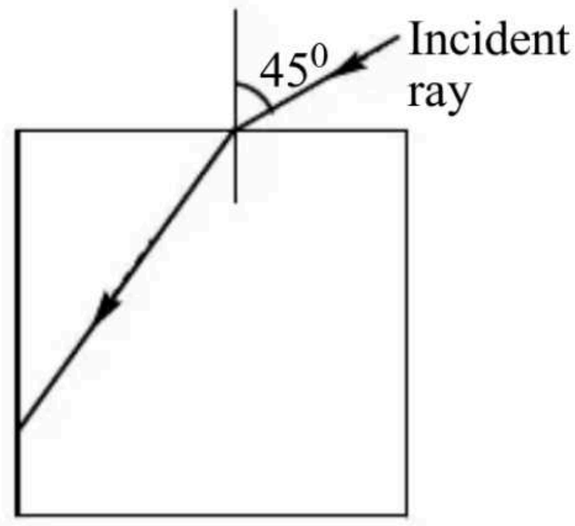
129. The given graph represents V-I characteristic for a semiconductor device.
Which of the following statements is correct ?



(1) It is V – I characteristic for solar cell where, point A represents open circuit voltage and point B short circuit current
(2) It is for a solar cell and point A and B represent open circuit voltage and current, respectively.
(3) It is for a photodiode and points A and B represents open circuit voltage and current, respectively
(4) It is for a LED and points A and B represent open circuit voltage and short circuit current, respectively.

130. In forward biasing of the p-n junction
(1) The positive terminal of the battery is connected to p-side and the depletion region becomes thick
(2) The positive terminal of the battery is connected to n-side and the depletion region becomes thin
(3) The positive terminal of the battery is connected to n-side and the depletion region becomes thick
(4) The positive terminal of the battery is connected to p-side and the depletion region becomes thin.

131. For the given incident ray as shown in figure, for the condition of total internal reflection at vertical face of the slab the required refractive index of the slab will be



(1) $\frac{\sqrt{3}+1}{2}$
(2) $\frac{\sqrt{2}+1}{2}$
(3) $\sqrt{\frac{3}{2}}$
(4) $\sqrt{\frac{7}{6}}$

132. A step-up transformer operates on a 230 V line and supplies a load of 2 ampere. The ratio of the primary and secondary windings is 1 : 25. The current in the primary is

(1) 15A

(2) 50A

(3) 25 A

(4) 12.5 A

133. Two bar magnets having same geometry with magnetic moments M and 2M, are first placed in such a way that their similar poles are same side then its time period of oscillations is T_1. Now the polarity of one of the magnet is reversed then time period of oscillation is T_2, then

(1) $T_1 < T_2$

(2) $T_1 = T_2$

(3) $T_1 > T_2$

(4) $T_2 = \infty$

134. The masses of the wires of copper is in the ratio of 1 : 3 : 5 and their lengths are in the ratio of 5 : 3 : 1. The ratio of their electrical resistance is

(1) 1 : 3 : 5

(2) 5 : 3 : 1

(3) 1 : 25 : 125

(4) 125 : 15 : 1

135. A 10 eV electron is circulating in a plane at right angles to a uniform field of magnetic induction $10^{-4} Wb/m^2$ (=1.0 gauss). The orbital radius of electron is

(1) 11 cm

(2) 18 cm

(3) 12 cm

(4) 16 cm

136. A gaseous mixture contains oxygen and nitrogen in the ratio of 1:4 by weight. Therefore, the ratio of their number of molecules is

(1) 1:4

(2) 1:8

(3) 7:32

(4) 3:16

137. For the redox reaction

$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$

The correct coefficient of the reactants for the balanced reaction are

	MnO_4^-	$C_2O_4^{2-}$	H^+
(1)	2	5	16
(2)	16	5	2
(3)	5	16	2
(4)	2	16	5

138. Number of electrons possible with n=3 and m=0 are

(1) 6

(2) 8

(3) 2

(4) 4

139. Incorrect order of following is

(1) $Na_2O > MgO > Al_2O_3 > SiO_2$..... basic nature

(2) Cl>P>S>Si……IP_1

(3) O > N > F > C…..IP_2

(4) F>O>N>C……EN

140. The ionic radii (in $\overset{o}{A}$) of N^{3-}, O^{2-} and F^- respectively are

(1) 1.36, 1.40 and 1.71

(2) 1.36, 1.71 and 1.40

(3) 1.71, 1.40 and 1.36

(4) 1.71, 1.36 and 1.40

141. Amongst H_2O, H_2S, H_2Se and H_2Te, the one with the highest boiling point is

(1) H_2O because of hydrogen bonding

(2) H_2Te because of higher molecular weight

(3) H_2S because of hydrogen bonding

(4) H_2Se because of lower molecular weight

142. Arrange the following compounds in order of increasing dipole moment,

toluene (I),

m-dichlorobenzene (II),

o-dichlorobenzene (III),

p-dichlorobenzene (IV)

(1) II < IV < I < III

(2) IV < I < II < III

(3) IV <II < III < I

(4) II < I < III<IV

143. Assuming Z-axis as the internuclear axis which of the following is a zero overlap

(1) s-s

(2) s-p_y

(3) p_z-p_z

(4) p_x-p_x

144. The rate of diffusion of methane at a given temperature is twice that of a gas *X*. The molecular weight of *X* is

(1) 64.0

(2) 32.0

(3) 4.0

(4) 8.0

145. For reaction ,

$CO(g)+H_2O(g) \rightleftharpoons CO_2(g)+H_2(g)$,

At a given temperature, the equilibrium amount of $CO_2(g)$ can be increased by

(1) Adding a suitable catalyst

(2) Adding an inert gas

(3) Decreasing the volume of the container

(4) Increasing the amount of $CO(g)$

146. $Ag^+ + NH_3 \rightleftharpoons [Ag(NH_3)]^+ ; K_1 = 3.5\times10^{-3}$

$[Ag(NH_3)]^+ + NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ ; K_2 = 1.7\times10^{-3}$

then the formation constant of $[Ag(NH_3)_2]^+$ is

(1) 6.08×10^{-6}

(2) 6.08×10^{6}

(3) 6.08×10^{-9}

(4) None of these

147. Which of the following statements is false?

(1) Work is a state function

(2) Temperature is a state function

(3) Change in the state is completely defined when the initial and final states are specified

(4) Work appears at the boundary of the system

148. The difference between heats of reaction at constant pressure and constant volume for the reaction

$2C_6H_6(l)+15O_2 \rightarrow 12CO_2(g)+6H_2O(l)$

At 25^0C in kJ is

(1) -7.43

(2) +3.72

(3) -3.72

(4) +7.43

149. A 0.2M sucrose solution is isotonic with 0.08M $BaCl_2$ solution . Degree of ionization of $BaCl_2$ is

(1) 0.87

(2) 0.75

(3) 0.66

(4) 0.5

150. For which of the following battery, cell potential remains constant during its life time

(1) dry cell

(2) lead accumulator

(3) mercury cell

(4) nickel cadmium battery

151. The gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y^- and 1 M Z^- at 25^0C. If the order of reduction potential is $Z/Z^- > Y/Y^- > X^+/X$ then

(1) Y will oxidize X and not Z^-

(2) Y will oxidize Z^- and not X

(3) Y will oxidize both X and Z^-

(4) Y will reduce both X^+ and Z

152. The rate constant of a reaction depends on

(1) Temperature

(2) Initial concentration of the reaction

(3) Time of reaction

(4) Extent of reaction

153. The half-life period of a radioactive element is 140 days. After 560 days, one gram of the element will reduce to.

(1) $\frac{1}{2}g$

(2) $\frac{1}{4}g$

(3) $\frac{1}{8}g$

(4) $\frac{1}{16}g$

154. The temporary hardness of water is due to calcium bicarbonate can be removed by adding

(1) $CaCO_3$

(2) $Ca(OH)_2$

(3) $CaCl_2$

(4) HCl

155. Which one of the following IIA metal sulphates has its hydration enthalpy greater than its lattice enthalpy?

(1) $CaSO_4$

(2) $BeSO_4$

(3) $BaSO_4$

(4) $SrSO_4$

156. A cobalt salt in borax bead test gives

(1) blue coloured $Co(BO_2)_2$

(2) orange coloured $Co(BO_2)_2$

(3) green coloured $Co(BO_2)_2$

(4) yellow coloured $Co(BO_2)_2$

157. Which of the following halides is least stable and has doubtful existence?

(1) CCl_4

(2) GeI_4

(3) SnI_4

(4) PbI_4

158. Which of the following has $-O-O-$ linkage?

(1) $H_2S_2O_6$

(2) $H_2S_2O_8$

(3) $H_2S_2O_3$

(4) $H_2S_4O_6$

159. Which one of the following is the strongest base?

(1) AsH_3

(2) NH_3

(3) PH_3

(4) SbH_3

160. The formula of first noble gas compound is

(1) $XePtF_6$

(2) $XePtF_2$

(3) $XePtF_4$

(4) $XePtF_5$

161. The set with correct order of acidic strength is

(1) $HClO < HClO_2 < HClO_3 < HClO_4$

(2) $HClO_4 < HClO_3 < HClO_2 < HClO$

(3) $HClO < HClO_4 < HClO_3 < HClO_2$

(4) $HClO_4 < HClO_2 < HClO_3 < HClO$

162. Which of the following transition metal is present in misch metal

(1) Cu

(2) Fe

(3) Zn

(4) Mn

163. Which of the following complex absorbs light of maximum wavelength

(1) $\left[Co(H_2O)_6\right]^{+3}$

(2) $[CoF_6]^{-3}$

(3) $\left[Co(CN)_6\right]^{-3}$

(4) $\left[Co(NH_3)_6\right]^{+3}$

164. $\left[Co(NH_3)_6\right]\left[Cr(CN)_6\right]$ and $\left[Cr(NH_3)_6\right]\left[Co(CN)_6\right]$ are

(1) Linkage isomers

(2) Coordination position isomers

(3) Coordination isomers

(4) Ionisation isomers

165. De-penicillamine is used to remove excess of _____from plants / animal system

(1) Ti, Pb

(2) Mn, Cr

(3) Co, Ni

(4) Fe, Cu

166. How many optically active stereoisomers are possible for butane-2, 3-diol?

(1) 1

(2) 2

(3) 3

(4) 4

167. A 0.01moles of gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g of CO_2. The empirical formula of the hydrocarbon is

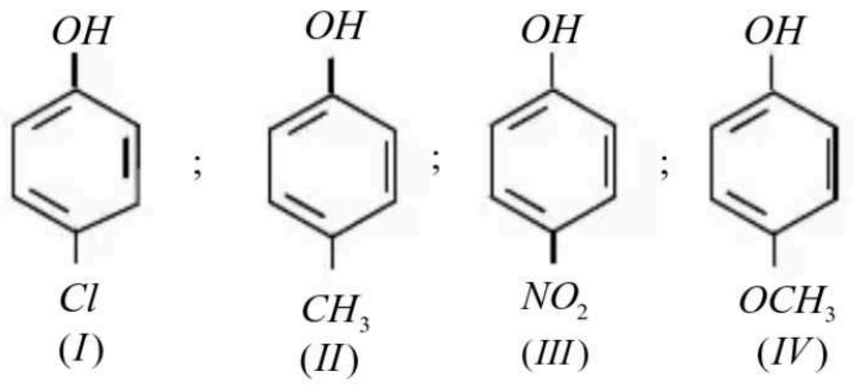
(1) C_2H_4

(2) C_3H_4

(3) C_6H_5

(4) C_7H_8

168. Arrange the following compounds in the order of decreasing acidity



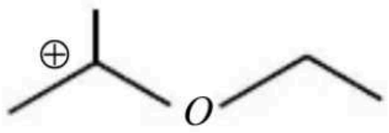
(1) II > IV > I > III

(2) I > II > III > IV

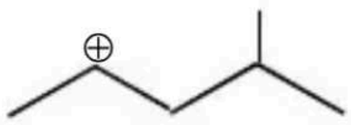
(3) III > I > II > IV

(4) IV > III > I > II

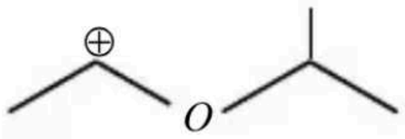
169. The correct stability order for the following species is



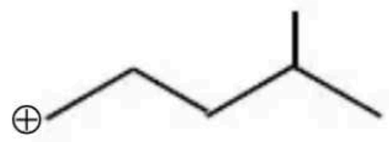
(I)



(II)



(III)



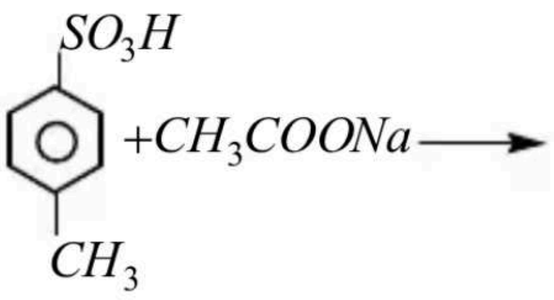
(IV)

(1) (II) > (IV) > (I) > (III)

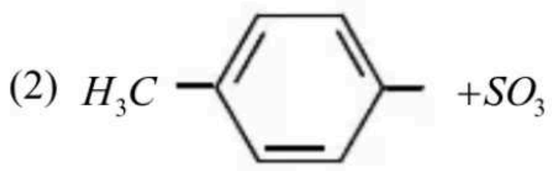
(2) (I) > (II) > (III) > (IV)

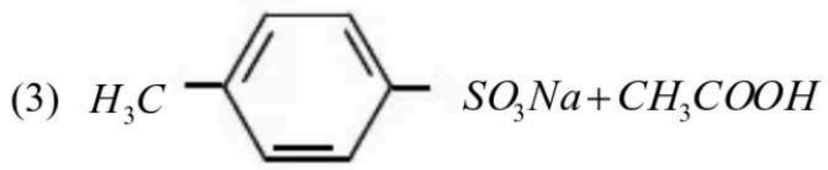
(3) (II) > (I) > (IV) > (III)

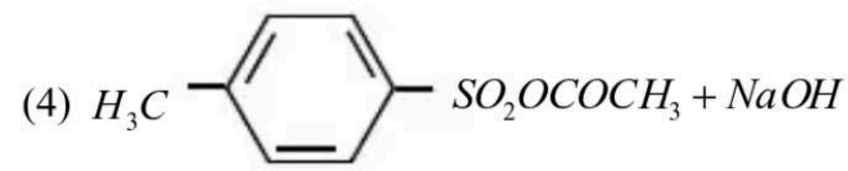
(4) (I) > (III) > (II) > (IV)

170. 

(1) 

(2) 

(3) 

(4) 

171. $(CH_3)_3CMgCl$ on reaction with D_2O produces

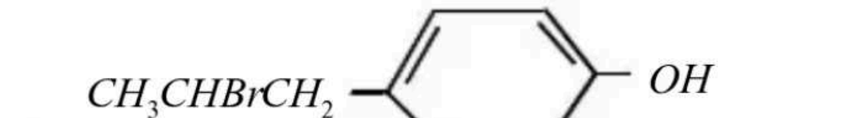
(1) $(CH_3)_3CD$

(2) $(CH_3)_3COD$

(3) $(CD_3)_3CD$

(4) $(CD_3)_3COD$

172. The reagent(s) for the following conversion,



is/are

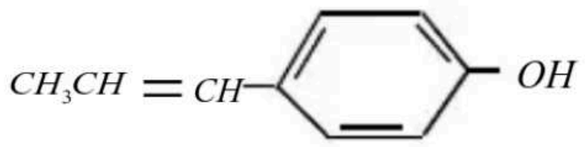
(1) Alcoholic KOH

(2) Alcoholic KOH followed by $NaNH_2$

(3) Aqueous KOH followed by $NaNH_2$

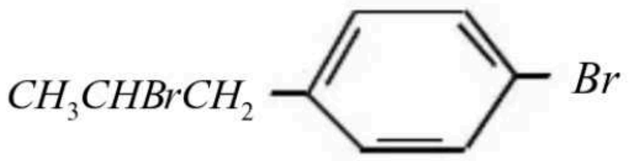
(4) Zn/CH_3OH

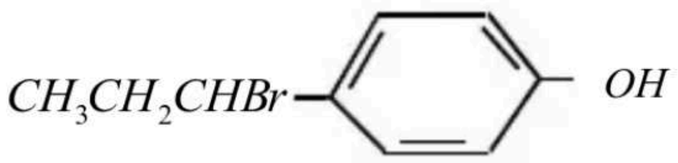
173. The reaction of

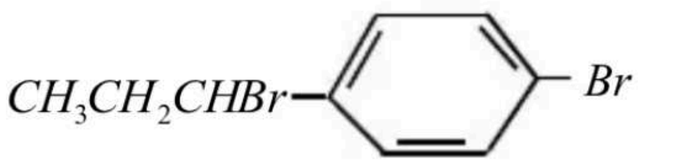


with HBr gives

(1)

(2) 

(3) 

(4) 

174. The reaction condition leading to the best yield of C_2H_5Cl is

(1) $C_2H_6(excess)+Cl_2 \xrightarrow{UVlight}$

(2) $C_2H_6+Cl_2(excess) \xrightarrow[roomtemp.]{drak}$

(3) $C_2H_6+Cl_2(excess) \xrightarrow{UVlight}$

(4) $C_2H_6+Cl_2 \xrightarrow{CCl_4}$

175. In the reaction,

$$CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{Alc.KOH} C$$

The product C is

(1) Acetaldehyde

(2) Acetylene

(3) Ethylene

(4) Acetyl chloride

176. Which of the following product is obtained when acetamide is treated with $LiAlH_4$

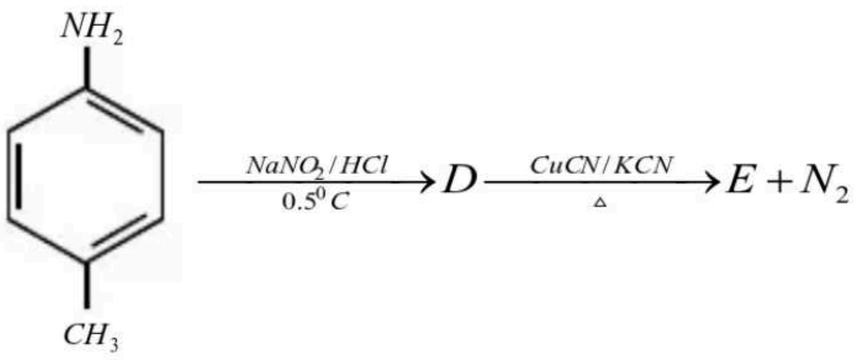
(1) methanamide

(2) ethanoicacid

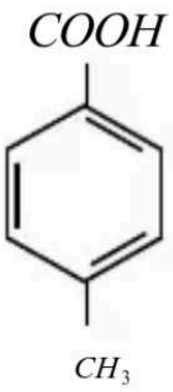
(3) methanamine

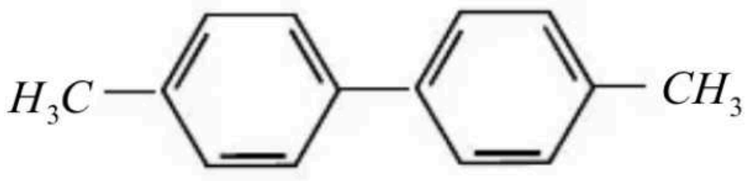
(4) ethanamine

177. In the reaction,

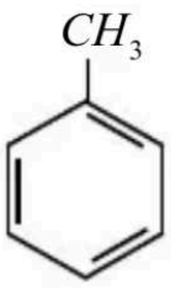


The product E is

(1) 

(2) 

(3) 

(4) 

178. In the following sequence of reactions

$$\text{Toluene} \xrightarrow{KMnO_4} A \xrightarrow{SOCl_2} B \xrightarrow[BaSO_4]{H_2/Pd} C,$$

The product C is

(1) C_6H_5COOH

(2) $C_6H_5CH_3$

(3) $C_6H_5CH_2OH$

(4) C_6H_5CHO

179. Glucose can react with

(1) HCN

(2) Schiff's reagent

(3) NH_3

(4) $NaHSO_3$

180. Which of the following is the solvent used in transparent soaps

(1) Benzene

(2) Water

(3) Ethanol

(4) Ether

SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI

SR MEDICAL ALL & LTC-IC

Date : 27.04.20

NEET GRAND TEST – 06 KEY

BOT:- 1,2,3,7,9,11,13,14,17,19,21,23,25,26,29,31,33,34,37,39,41,42,45,47,49,51,53,55,56,59,61, 63,65,67,69,71,72,75,76,79,81,863,85,87,89

ZOO:- 4,5,6,8,10,12,15,16,18,20,22,24,27,28,30,32,35,36,38,40,43,44,46,48,50,52,54,57,58,60, 62,64,66,68,70,73,74,77,78,80,82,84,86,88,90

BIOLOGY

1) 3	2) 1	3) 2	4) 3	5) 4	6) 3	7) 1	8) 1	9) 4	10) 3
11) 1	12) 4	13) 2	14) 1	15) 2	16) 3	17) 2	18) 2	19) 3	20) 4
21) 4	22) 4	23) 4	24) 4	25) 2	26) 3	27) 2	28) 2	29) 4	30) 4
31) 4	32) 3	33) 1	34) 3	35) 1	36) 4	37) 4	38) 2	39) 2	40) 2
41) 3	42) 2	43) 2	44) 3	45) 4	46) 4	47) 4	48) 4	49) 4	50) 2
51) 4	52) 4	53) 4	54) 4	55) 3	56) 2	57) 3	58) 2	59) 3	60) 4
61) 4	62) 4	63) 3	64) 4	65) 3	66) 2	67) 4	68) 3	69) 3	70) 3
71) 4	72) 1	73) 2	74) 4	75) 3	76) 2	77) 4	78) 2	79) 3	80) 4
81) 1	82) 2	83) 1	84) 3	85) 4	86) 1	87) 3	88) 3	89) 3	90) 1

PHYSICS

91) 4	92) 1	93) 1	94) 4	95) 2	96) 2	97) 1	98) 3	99) 4	100) 3
101) 3	102) 3	103) 1	104) 4	105) 3	106) 3	107) 3	108) 1	109) 1	110) 4
111) 3	112) 2	113) 4	114) 2	115) 3	116) 3	117) 3	118) 2	119) 2	120) 2
121) 2	122) 1	123) 4	124) 1	125) 1	126) 1	127) 3	128) 4	129) 1	130) 4
131) 3	132) 2	133) 1	134) 4	135) 1					

CHEMISTRY

136) 3	137) 1	138) 1	139) 3	140) 3	141) 1	142) 2	143) 2	144) 1	145) 4
146) 1	147) 1	148) 1	149) 2	150) 3	151) 1	152) 1	153) 4	154) 2	155) 2
156) 1	157) 4	158) 2	159) 2	160) 1	161) 1	162) 2	163) 2	164) 3	165) 4
166) 2	167) 4	168) 3	169) 4	170) 3	171) 1	172) 2	173) 3	174) 1	175) 3
176) 4	177) 3	178) 4	179) 1	180) 3					

SOLUTIONS_27-04-2020

PHYSICS

91. Diameter of the ball

= MSR + CSR ×(Least count) – Zero error

= 5 mm + 25 × 0.001 cm – (– 0.004) cm

= 0.5 cm + 25×0.001 cm – (–0.004) cm = 0.529 cm.

92. Given : $x = 9t^2 - t^3$

Speed $v = \frac{dx}{dt} = \frac{d}{dt}\left(9t^2 - t^3\right) = 18t - 3t^2$.

For maximum speed,

$$\frac{dv}{dt} = 0 \Rightarrow 18 - 6t = 0$$

$$\therefore t = 3s.$$

$$\therefore x_{\max} = 81m - 27m = 54m.$$

(From $x = 9t^2 - t^3$)

93. Two vectors $\vec{A}$ and $\vec{B}$ are orthogonal to each other, if their scalar product is zero

$i.e. \vec{A}.\vec{B} = 0.$

Here, $\vec{A} = \cos\omega t\hat{i} + \sin\omega t\hat{j}$ and

$$\vec{B} = \cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}$$

$$\therefore \vec{A}.\vec{B} = \left(\cos\omega t\hat{i} + \sin\omega t\hat{j}\right)$$

$$\left(\cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}\right)$$

$$= \cos\omega t\cos\frac{\omega t}{2} + \sin\omega t\sin\frac{\omega t}{2}$$

$$\left(\because \hat{i}.\hat{i} = \hat{j}.\hat{j} = 1 and \hat{i}.\hat{j} = \hat{j}.\hat{i} = 0\right)$$

$$= \cos\left(\omega t - \frac{\omega t}{2}\right)$$

$$\left(\because \cos(A - B) = \cos A\cos B + \sin A\sin B\right)$$

But $\vec{A}.\vec{B} = 0$ (as $\vec{A}$ and $\vec{B}$ are orthogonal to each other)

$$\therefore \cos\left(\omega t - \frac{\omega t}{2}\right) = 0$$

$$\cos\left(\omega t - \frac{\omega t}{2}\right) = \cos\frac{\pi}{2} \text{ or } \omega t - \frac{\omega t}{2} = \frac{\pi}{2}$$

$$\frac{\omega t}{2} = \frac{\pi}{2} \text{ or } \frac{\pi}{\omega}$$

94. For vertical equilibrium on the road,

$$N\cos\theta = mg + f\sin\theta$$

$$mg = N\cos\theta - f\sin\theta$$

Centripetal force for safe turning,

$$N\sin\theta + f\cos\theta = \frac{mv^2}{R}$$

From eqns. (i) and (ii), we get

$$\frac{v^2}{Rg} = \frac{N\sin\theta + f\cos\theta}{N\cos\theta - f\sin\theta}$$

$$\Rightarrow \frac{v_{\max}^2}{Rg} = \frac{N\sin\theta + \mu_s N\cos\theta}{N\cos\theta - \mu_s N\sin\theta}$$

$$v_{\max} = \sqrt{Rg\left(\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)}$$

95. Rate of burning of fuel $\left(\frac{dm}{dt}\right) = 1kg/s$ and velocity of ejected fuel (v) = 60 km/s $= 60 \times 10^3 m/s$

Force = Rate of change of momentum

$$= \frac{dp}{dt} = \frac{d(mv)}{dt} = v\frac{dm}{dt} = \left(60 \times 10^3\right) \times 1 = 60000N \cdot$$

96. Here, $U = \frac{A}{r^2} - \frac{B}{r}$

For equilibrium, $\frac{dU}{dr} = 0$

$$\therefore -\frac{2A}{r^3} + \frac{B}{r^2} = 0 \text{ or } \frac{2A}{r^3} = \frac{B}{r^2} \text{ or } r = \frac{2A}{B}$$

For stable equilibrium, $\frac{d^2U}{dr^2} > 0$

$$\frac{d^2U}{dr^2} = \frac{6A}{r^4} - \frac{2B}{r^3}$$

$$\left.\frac{d^2U}{dr^2}\right|_{r=(2A/B)} = \frac{6AB^4}{16A^4} - \frac{2B^4}{8A^3} = \frac{B^4}{8A^3} > 0$$

So for stable equilibrium, the distance of the particle is $\frac{2A}{B}$.

97. Work done required to bring a object to rest

$$\Delta W = \Delta KE$$

$\Delta W = \frac{1}{2}I\omega^2$; where I = moment of inertia

For same $\omega, \Delta W \propto I$

For a solid sphere, $I_A = \frac{2}{5}MR^2$

For disc $I_B = \frac{MR^2}{2}$

For a circular ring, $I_C = MR^2$

$\therefore I_C > I_B > I_A \ \therefore W_C > W_B > W_A$

98. As the system is initially at rest, therefore, initial angular momentum $L_i = 0$.

According to the principle of conservation of angular momentum, final angular momentum, $L_f = 0$.

$\therefore$ Angular momentum of disc = Angular momentum of man is in opposite direction of platform.

i.e., $mvR = I\omega$

or $\omega = \frac{m\upsilon R}{I} = \frac{50 \times 1 \times 2}{200} = \frac{1}{2} rads^{-1}$

Angular velocity of man relative to platform is

$\omega_r = \omega + \frac{\upsilon}{R} = \frac{1}{2} + \frac{1}{2} = 1 rads^{-1}$

Time taken by the man to complete one revolution is $T = \frac{2\pi}{\omega_r} = \frac{2\pi}{1} = 2\pi s$

99. If universal Gravitational constant becomes ten times, then $G' = 10G$.

So, acceleration due to gravity increases. i.e, (d) is wrong option.

100. From Wien's law, $\lambda_{\max} T$ =constant

So, $\lambda_{\max_1} T_1 = \lambda_{\max_2} T_2$

$\Rightarrow \lambda_0 T = \frac{3\lambda_0}{4} T' \Rightarrow \frac{T'}{T} = \frac{4}{3}$

According to Stefan –Boltzman law, energy emitted unit time by a black body is $Ae\sigma T^4$, i.e., power radiated.

$\therefore P \propto T^4$

So, $\frac{P_2}{P_1} = \left(\frac{T'}{T}\right)^4 \Rightarrow n = \left(\frac{4}{3}\right)^4 = \frac{256}{81}$

101. Young's modulus,

$Y = \frac{FL}{A\Delta L} = \frac{4FL}{\pi D^2 \Delta L}$ or $\Delta L = \frac{4FL}{\pi D^2 Y}$

Where F is the force applied, L is the length, D is the diameter and ΔL is the extension of the wire respectively. As each wire is made up of same material therefore their Young's modulus is same for each wire.

For all the four wires, Y, F(= tension) are the same.

$\therefore \Delta L \propto \frac{L}{D^2}$

In (a) $\frac{L}{D^2} = \frac{200cm}{(0.2cm)^2} = 5 \times 10^3 cm^{-1}$

In (b) $\frac{L}{D^2} = \frac{300cm}{(0.3cm)^2} = 3.3 \times 10^3 cm^{-1}$

In (c) $\frac{L}{D^2} = \frac{50cm}{(0.05cm)^2} = 20 \times 10^3 cm^{-1}$

In (d) $\frac{L}{D^2} = \frac{100cm}{(0.1cm)^2} = 10 \times 10^3 cm^{-1}$

Hence, ΔL is maximum in (c)

102. The internal energy of 2 moles of O_2 atom is

$U_{O_2} = \frac{n_1 f_1}{2} RT = 2 \times \frac{5}{2} \times RT$

$U_{O_2} = 5RT$

The internal energy of 4 moles of Ar atom is

$U_{Ar} = \frac{n_2 f_2 RT}{2} = 4 \times \frac{3}{2} \times RT = 6RT$

$\therefore$ The total internal energy of the system is

$U_{Ar} = \frac{n_2 f_2 RT}{2} = 4 \times \frac{3}{2} \times RT = 6RT$

$\therefore$ The total internal energy of the system is $U = U_{O_2} + U_{Ar} = 5RT + 6RT = 11RT$

103. Conceptual

104. Heat required to melt 1 kg ice at 0ºC to water at 0ºC is

$Q = m_{ice} L_{ice} = (1kg)(80cal/g)$

$= (1000g)(80cal/g) = 8 \times 10^4 cal$

Change in entropy,

$\Delta S = \frac{Q}{T} = \frac{8 \times 10^4 cal}{(273K)}$

= 293 cal/K

105. $y = \sin\omega t - \cos\omega t$

$$= \sqrt{2}\left[\frac{1}{\sqrt{2}}\sin\omega t - \frac{1}{\sqrt{2}}\cos\omega t\right] = \sqrt{2}\sin\left(\omega t - \frac{\pi}{4}\right)$$

It represents a SHM with time period, $T = \frac{2\pi}{\omega}$.

$$y = 1 + \omega t + \omega^2 t^2$$

It represents a non-periodic motion. Also it is not physically acceptable as $y \to \infty$ as $t \to \infty$.

106. For simple harmonic motion velocity $v = \omega\sqrt{a^2 - x^2}$ at displacement x.

$$10 = \omega\sqrt{a^2 - 16} \quad \text{____ (i)}$$

$$8 = \omega\sqrt{a^2 - 25} \quad \text{____ (ii)}$$

$$\frac{100}{\omega^2} = a^2 - 16 \quad \text{_____ (iii)}$$

$$\frac{64}{\omega^2} = a^2 - 25 \quad \text{_____ (iv)}$$

$\therefore$ Equation (iii) – (iv) gives $\frac{36}{\omega^2} = 9$

$$\Rightarrow \omega = 2rad/s$$

or $T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi\sec$

107. The required frequency of sound heard by the driver of second car is given as

$$\upsilon' = \upsilon\left(\frac{v + v_0}{v - v_s}\right),$$

Where v = velocity of sound

v_0 =velocity of observer, i.e., second car

v_s = velocity of source i.e., first car

$$\upsilon' = 400\left(\frac{340 + 16.5}{340 - 22}\right) = 400\left(\frac{356.5}{318}\right)$$

$$\upsilon' \approx 448Hz$$

108. $y = 0.25\sin(10\pi x - 2\pi t)$

$$y_{\max} = 0.25$$

$$k = \frac{2\pi}{\lambda} = 10\pi \Rightarrow \lambda = 0.2m$$

$$\omega = 2\pi f = 2\pi \Rightarrow f = 1Hz$$

The sign is negative inside the bracket. Therefore this wave travels in the positive x-direction.

109. For isolated capacitor, charge Q = constant.

Electrostatic force, $F_{plate} = \frac{Q^2}{2A\varepsilon_0}$ F

is independent of the distance between plates.

110. According to the given problem, $U_s = U_p$

Substituting the values of U_s and U_p from equations (ii) and (iv), we get

$$\frac{1}{2}\frac{C_1}{n_1}(4V)^2 = \frac{1}{2}(n_2C_2)(V)^2$$

or $\frac{C_1 16}{n_1} = n_2C_2$ or $C_2 = \frac{16C_1}{n_1n_2}$

111. $R = \frac{V}{i_g} - G, in\ series$

112. $(47 \pm 4.7)k\Omega = 47 \times 10^3 \pm 10\%\Omega$

$\therefore$ Yellow – Violet – Orange – Silver

113. Here, Length of each rod, $l = 1\text{m}$

Area of cross-section of each rod,

$$A = 0.01cm^2 = 0.01 \times 10^{-4}m^2$$

Resistivity of copper rod,

$$\rho_{cu} = 1.7 \times 10^{-6}\Omega cm$$

$$= 1.7 \times 10^{-6} \times 10^{-2}\Omega m = 1.7 \times 10^{-8}\Omega m$$

Resistivity of iron rod,

$$\rho_{Fe} = 10^{-5}\Omega cm$$

$$= 10^{-5} \times 10^{-2}\Omega m = 10^{-7}\Omega m$$

$\therefore$ Resistance of copper rod,

$$R_{Cu} = \rho_{Fe}\frac{l}{A}$$

And resistance of iron rod, $R_{Fe} = \rho_{Fe}\frac{l}{A}$

As copper and iron rods are connected in series, therefore equivalent resistance is

$$R = R_{Cu} + R_{Fe} = \rho_{Cu}\frac{l}{A} + \rho_{Fe}\frac{l}{A} = (\rho_{Cu} + \rho_{Fe})\frac{l}{A}$$

Voltage required to produce 1 A current in the rods is $V = IR = (1)(R_{Cu} + R_{Fe})$

$$= (\rho_{Cu} + \rho_{Fe})\left(\frac{l}{A}\right)$$

$$= \left(1.7 \times 10^{-8} + 10^{-7}\right)\left(\frac{1}{0.01 \times 10^{-4}}\right)V$$

$$=10^{-7}\left(0.17+1\right)\left(10^{6}\right)V=1.17\times10^{-1}V=0.117V$$

114. $$K_\alpha = K_P\left(\frac{q_\alpha}{q_p}\right)^2\left(\frac{m_p}{m_\alpha}\right)\left(\frac{R_\alpha}{R_p}\right)^2$$

Here, $K_p = 1MeV, \frac{q_\alpha}{q_p}=2, \frac{m_p}{m_\alpha}=\frac{1}{4}$

and $\frac{R_\alpha}{R_p}=1$

$$\therefore K_\alpha = (1MeV)(2)^2\left(\frac{1}{4}\right)(1)^2 = 1MeV$$

115. Magnetic moment of the loop.

$$M = NIA = 2000\times2\times1.5\times10^{-4} = 0.6J/T$$

Torque $\tau = MB\sin30^\circ$

$$=0.6\times5\times10^{-2}\times\frac{1}{2}=1.5\times10^{-2}Nm$$

116. Magnetic susceptibility is negative for diamagnetic material only.

117. Quality factor of an L-C-R circuit is given by.

$$Q=\frac{1}{R}\sqrt{\frac{L}{C}}$$

118. As, $|\varepsilon| = M\frac{dI}{dt}$

$$=M\frac{d}{dt}\left(I_0\sin\omega t\right)=MI_0\omega\cos\omega t$$

$$\therefore \varepsilon_{\max}=0.005\times10\times100\pi\times1=5\pi$$

119. X-rays are used for the investigation of structure of solids.

120. Here, $\frac{I_1}{I_2}=n$

$$\frac{I_{\max}}{I_{\min}}=\left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2=\left(\frac{\sqrt{I_1/I_2}+1}{\sqrt{I_1/I_2}-1}\right)^2$$

$$=\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)^2$$

$$\frac{I_{\max}-I_{\min}}{I_{\max}+I_{\min}}=\frac{\frac{I_{\max}}{I_{\min}}-1}{\frac{I_{\max}}{I_{\min}}+1}$$

$$=\frac{\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)^2-1}{\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)^2+1}=\frac{\left(\sqrt{n}+1\right)^2-\left(\sqrt{n}-1\right)^2}{\left(\sqrt{n}+1\right)^2+\left(\sqrt{n}-1\right)^2}$$

$$=\frac{4\sqrt{n}}{2(n+1)}=\frac{2\sqrt{n}}{n+1}$$

121. Fringe width, $\beta=\frac{\lambda D}{d}$

Where D is the distance between slits and screen and d is the distance between the slits.

When D is doubled and d is reduced to half, then fringe width becomes

$$\beta'=\frac{\lambda(2D)}{(d/2)}=\frac{4\lambda D}{d}=4\beta$$

122. $A=60^\circ, \mu=\sqrt{3}, \delta_m=?$

$$\mu=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}} \quad \therefore \delta_m=60^\circ$$

123. Direction of propagation is parallel to $\vec{E}\times\vec{B}$

124. de Brogile wavelength associated with an electron is

$$\lambda=\frac{h}{P} \text{ or } P=\frac{h}{\lambda}$$

$$\therefore\frac{\Delta P}{P}=-\frac{\Delta\lambda}{\lambda}$$

$$\frac{P}{P_{initial}}=\frac{0.5}{100}$$

$$P_{initial}=200P$$

125. The work function has no effect on photoelectric current so long as $h\upsilon > W_0$. The photoelectric current proportional to the intensity of incident light. Since there is no change in the intensity of light, hence $I_1 = I_2$.

126. According to Rydberg formula

$$\frac{1}{\lambda}=R\left[\frac{1}{n_f^2}-\frac{1}{n_i^2}\right]$$

Here, $n_f=1, n_i=5$

$$\therefore\frac{1}{\lambda}=R\left[\frac{1}{1^2}-\frac{1}{5^2}\right]=R\left[\frac{1}{1}-\frac{1}{25}\right]=\frac{24}{25}R$$

According to conservation of linear momentum, we get

Momentum of photon = Moment of atom

$$\frac{h}{\lambda} = m\upsilon \text{ or}$$

$$\upsilon = \frac{h}{m\lambda} = \frac{h}{m}\left(\frac{24R}{25}\right) = \frac{24hR}{25m}$$

127. Given : $\lambda_A = 5\lambda, \lambda_B = \lambda$

At $t = 0, \left(N_0\right)_A = \left(N_0\right)_B$

$$\frac{N_A}{N_B} = \left(\frac{1}{e}\right)^2$$

According to radioactive decay, $\frac{N}{N_0} = e^{-\lambda t}$

$$\therefore \frac{N_A}{\left(N_0\right)_A} = e^{-\lambda_A t} \quad \ldots\ldots\ldots \text{(i)}$$

$$\frac{N_B}{\left(N_0\right)_B} = e^{-\lambda_B t} \quad \ldots\ldots\ldots \text{(ii)}$$

Divide (i) by (ii), we get

$$\frac{N_A}{N_B} = e^{-(\lambda_A - \lambda_B)t} \quad \text{or,} \quad \frac{N_A}{N_B} = e^{-(5\lambda - \lambda)t}$$

$$\text{or,} \left(\frac{1}{e}\right)^2 = e^{-4\lambda t} \text{ or,} \left(\frac{1}{e}\right)^2 = \left(\frac{1}{e}\right)^{4\lambda t}$$

$$\text{or,} \ 4\lambda t = 2 \Rightarrow t = \frac{2}{4\lambda} = \frac{1}{2\lambda}$$

128. Given $V_{BE} = 0; V_{CE} = 0$

$$\therefore \text{ Collecter current, } I_C = \frac{(20-0)}{4\times10^3}$$

$$\text{or } I_C = 5\times10^{-3}\,A = 5mA$$

Input voltage, $V_i = V_{BE} + I_B R_B$

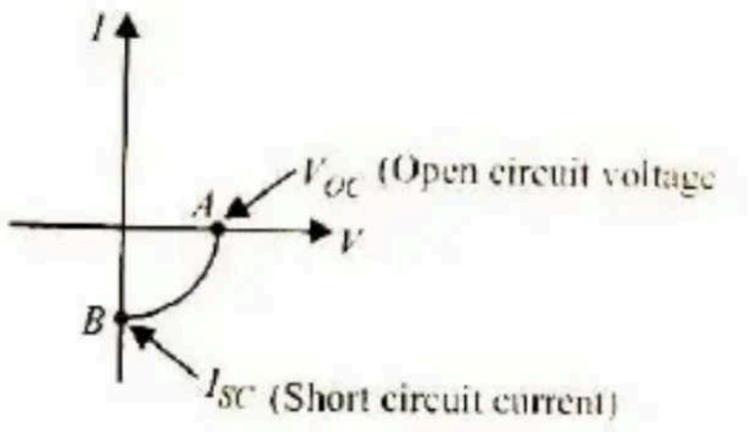
$$\text{or } V_i = 0 + I_B R_B or 20 = I_B \times 500 \times 10^3$$

$$\therefore I_B = \frac{20}{500\times10^3} = 40\mu A$$

$\therefore$ current gain,

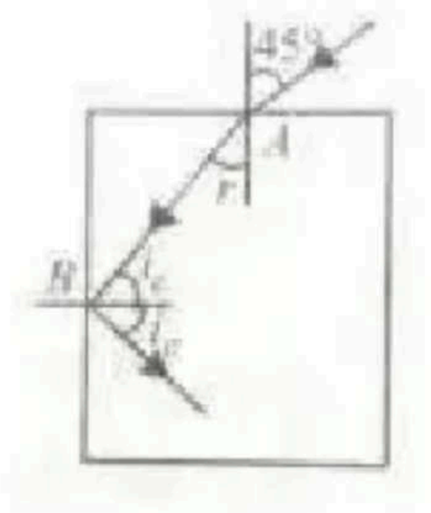
$$\beta = \frac{I_C}{I_B} = \frac{5\times10^{-3}}{40\times10^{-6}} = 125$$

129. The V-I characteristic for a solar cell is as shown the figure.



130. In forward biasing, the positive terminal of the battery is connected to p-side junction, the forward bais voltage opposes the potential barrier. Due to it, the depletion region becomes thin.

131. Applying Snell's law of refraction at A, we get



$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 45^\circ}{\sin r}$$

$$\therefore \sin r = 1/\sqrt{2}\mu$$

$$\therefore r = \sin^{-1}\left(\frac{1}{\sqrt{2}\mu}\right)$$

Applying the condition of total internal reflection at B, we get

$$i_c = \sin^{-1}\left(1/\mu\right)$$

Where i_c is the critical angle

Now, $r + i_c = 90^\circ = \pi/2$

$$\therefore \sin^{-1}\frac{1}{\sqrt{2}\mu} = \frac{\pi}{2} - \sin^{-1}\frac{1}{\mu}$$

$$\text{or , } \sin^{-1}\frac{1}{\sqrt{2}\mu} = \cos^{-1}\frac{1}{\mu}$$

$$\therefore \frac{1}{\sqrt{2}\mu} = \frac{\sqrt{\mu^2 - 1}}{\mu} \text{ or } \frac{1}{2} = \mu^2 - 1.$$

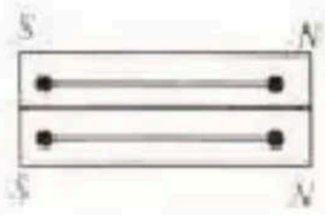
$$\therefore \mu = \sqrt{3/2}$$

132. $\frac{E_P}{E_S} = \frac{N_P}{N_S} = \frac{1}{25}; E_S = 5750V$

$$\frac{E_P}{E_S} = \frac{N_P}{N_S} = \frac{1}{25}; E_S = 5750V$$

$$I_P = \frac{P_P}{V_P} = \frac{P_S}{V_P} = \frac{2 \times 5750}{230} = 50A$$

133. a) S



i)

$(i) M = M_1 M_2$ $\qquad$ $(ii) M = M_1 - M_2$

$I = I_1 + I_2$ $\qquad$ $I = I_1 + I_2$

i) Similar poles are placed at the same side (sum position)

ii) Opposite poles are placed at the same side (difference position)

I_1 and I_2 are the moments of inertia of the magnets and M_1 and M_2 are the moments of the magnets. Here $M_1 = M$ and $M_2 = 2M, I_1 = I_2 = I$ (say), for same geometry.

$$\therefore T_1 = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1 + M_2)H}} = 2\pi\sqrt{\frac{2I}{(M + 2M)H}}$$

For same position.

And

$$T_2 = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1 + M_2)H}} = 2\pi\sqrt{\frac{2I}{(2M - M)H}}$$

For difference position.

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{M}{3M}} = \frac{1}{\sqrt{3}} < 1.$$

$$\therefore T_1 < T_2$$

134. $m = l \times area \times density$

$$R \propto = \frac{1}{Area} \propto \frac{l^2}{m}$$

$$R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3}$$

$$R_1 : R_2 : R_3 = \frac{25}{1} : \frac{9}{3} : \frac{1}{5} = 125 : 15 : 1$$

135. $r = \frac{mV}{Bq}$

CHEMISTRY

136. $W_{O_2} = 1$ $\qquad$ $W_{N_2} = 4$

$n_{O_2} = \frac{1}{32}$ $\qquad$ $n_{N_2} = \frac{4}{28}$

1 mole $\longrightarrow$ N

$\frac{1}{32}$ moles $\longrightarrow \frac{N}{32}$ molecule

1 mole $\longrightarrow$ N

$\frac{4}{28} \longrightarrow \frac{4N}{28}$

$\frac{\cancel{N}}{32} \times \frac{28}{4\cancel{N}} = 7:32$

137.

by 1×2

$$\overset{+7}{MnO_4^-} + \overset{+3}{C_2O_4^{2-}} + H^+ \longrightarrow \overset{+2}{Mn^{2+}} + \overset{+4}{CO_2} + H_2O$$

by 5

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

2, $\qquad$ 5, $\qquad$ 16

138. n = 3

m = 0

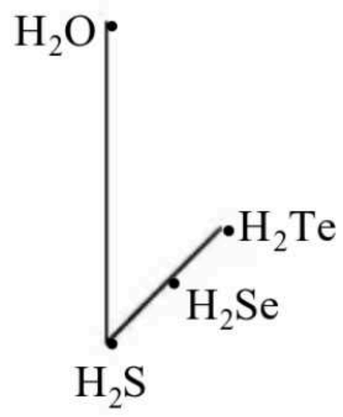
$3s, 3p_3, 3d_3^2 = \text{Total } 6e^-$

139. $C < N < O < F$

140. For isoelectronic species with increase in negative charge size increases

$N^{3-} > O^{2-} > F^{-}$

141.

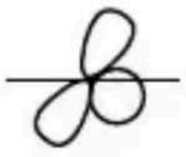


H_2O high B.P due to H.B

142. Disubstituted benzene

o > m > T > p

para has $\mu = 0$



143. Zero overlap

144. $r_{CH_4} = 2 \quad r_x = 1$

$M_{CH_4} = 16 \quad M_x = ?$

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

$$\frac{2}{1} = \sqrt{\frac{M_x}{16}} = M_x = 64$$

145. Addition of reactants shifts the equation to right side

146. $Ag^{+} + NH_3 \rightleftharpoons Ag(NH_3) + \; ; \; K_1 = 3.5 \times 10^{-3}$

$$\frac{[Ag(NH_3)^{+} + NH_3 \rightleftharpoons [Ag(NH_3)_2] \; ; K_2 = 1.7 \times 10^{-3}}{Ag^{+} + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^{+} \; ; \; K}$$

$K = K_1 \times K_2 = 3.5 \times 10^{-3} \times 1.7 \times 10^{-3} = 5.95 \times 10^{-6}$

147. Work $\rightarrow$ path function

148. $\Delta H - \Delta E = \Delta nRT = -3 \times 8.314 \times 10^{-3} \times 298 = -7.432\,KJ$

149. For isotonic solutions

$\pi_1 = \pi_2 \quad BaCl_2 \rightarrow Ba^{2+} + 2Cl^{-}$

$C_1 = C_2 \quad (1-\alpha) \quad \alpha + 2\alpha = 1 + 2\alpha$

$0.2\ M = (1 + 2\alpha)\ 0.08$

$$\frac{0.2}{0.08} = (1 + 2\alpha)$$

$(1 + 2\alpha) = 2.5$

$2\alpha = 1.5$

$\alpha = 0.75$

151. High SRP element can oxidise less SRP element $\boxed{SRP \propto O.N}$

152. Rate constant depends on T only

153. Number of half lives (x) $= \frac{t}{t_{1/2}} = \frac{560}{140} = 4$

$$N = \frac{N_0}{2^x}\frac{1}{2^4} = \frac{1}{16}$$

154. $Ca(OH_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3 + 2H_2O$

slaked lime is used in Clark's method.

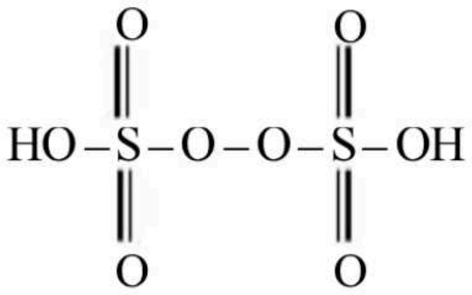
155. Size $\propto \frac{1}{\text{H.E}}$

For $BeSO_4$ H.E > L.E that is why it is more soluble then $BaSO_4$

156. $CoO + B_2O_3 \longrightarrow \underset{\text{blue bead}}{Co(BO_2)_2}$

157. PbI_4 is least stable due to inert pair effect

158. $H_2S_2O_8$



159. Basic nature of $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

160. $\overset{+1}{Xe}[\overset{+5}{Pt}F_6]$

Xenon hexa fluoro platinate (V)

161. Acidic strength increase with increase in oxidation state of chlorine

$HClO < HClO_2 < HClO_3 < HClO_4$

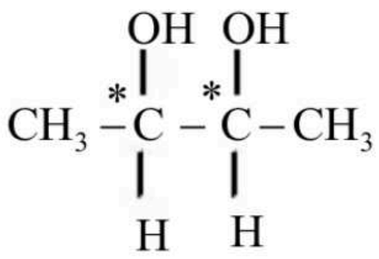
162. Misch metal is an alloy of Lanthanoid metal (95%) and Fe ($\simeq 5\%$) along with traces of S, C, Ca and Al.

163. Wave length of light absorbed is inversely proportional to strength of the ligand

164. Both are co-ordination isomers

165. De-pericillamine is used to remove Fe and Cu

166.



It has plane of symmetry.

Number of optically active isomers $= 2^{n-1} = 2^{2-1} = 2$

Number of optically in active isomers is mesomers $= 2^{n/2-1} = 2^{2/2-1} = 2^0 = 1$

167. $\underset{1\text{ mole}}{C_xH_y} + \left(x + \frac{y}{4}\right)O_2 \longrightarrow \underset{x\text{ mole}}{x\,CO_2} + \underset{y/2\text{ moles}}{\frac{y}{2}H_2O}$

0.01 moles $\longrightarrow$ 0.72 gm water

1 mole $\longrightarrow \frac{1}{0.01} \times 0.72 = 72$ gm

$= \frac{72}{18} = 4$ moles H_2O

$\therefore \frac{y}{2} = 4 \Rightarrow y = 8$

0.01 moles $\longrightarrow$ 3.08 gm CO_2

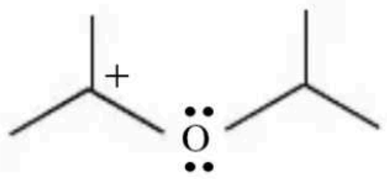
1 mole $\longrightarrow \frac{3.08}{0.01} = 308$ gm

$= \frac{308}{44} = 7$ moles CO_2

$\therefore$ Formula of Hydrocarbon is C_7H_8.

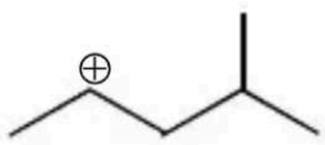
168. Presence of electron withdrawing group increase and electron releasing group decrease the acidic strength of phenols, alcohols and carboxylic acids.

169. Presence of lone pair atoms gives more stability to carbocation due to resonance



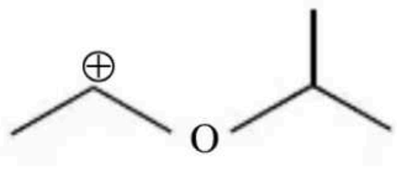
I

6 - Hyperconjugation and lone pair atom



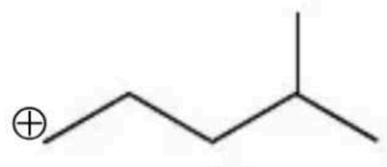
II

2° carbocation



III

3 - hyper conjugations and lone pair atom

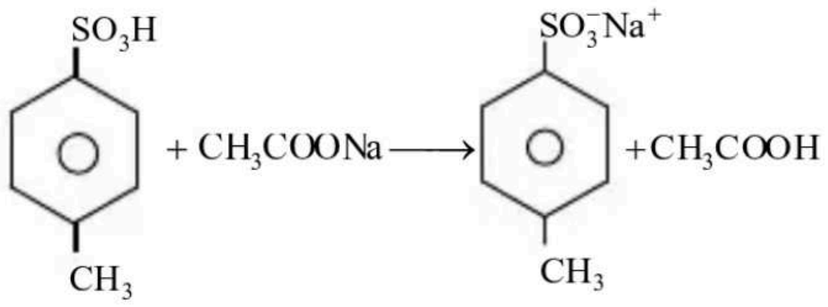


IV

1° carbocation

Stability of I > III > II > IV

170.



171.

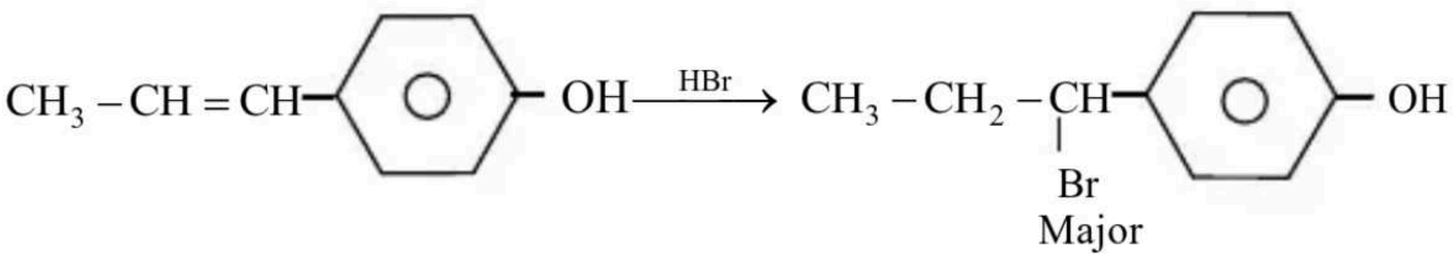
$$RMgX + D_2O \longrightarrow RD + Mg\begin{matrix} x \\ OD \end{matrix}$$

$$(CH_3)_3CMgCl + D_2O \longrightarrow (CH_3)_3C - D + Mg\begin{matrix} Cl \\ OD \end{matrix}$$

172.



173.

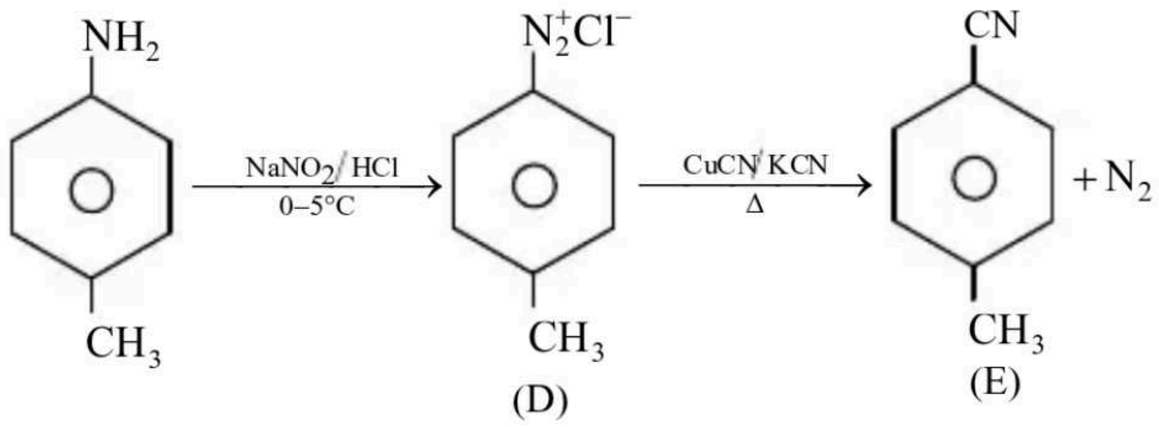


174. $C_2H_6 + Cl_2 \xrightarrow{UV} C_2H_5Cl + HCl$
excess

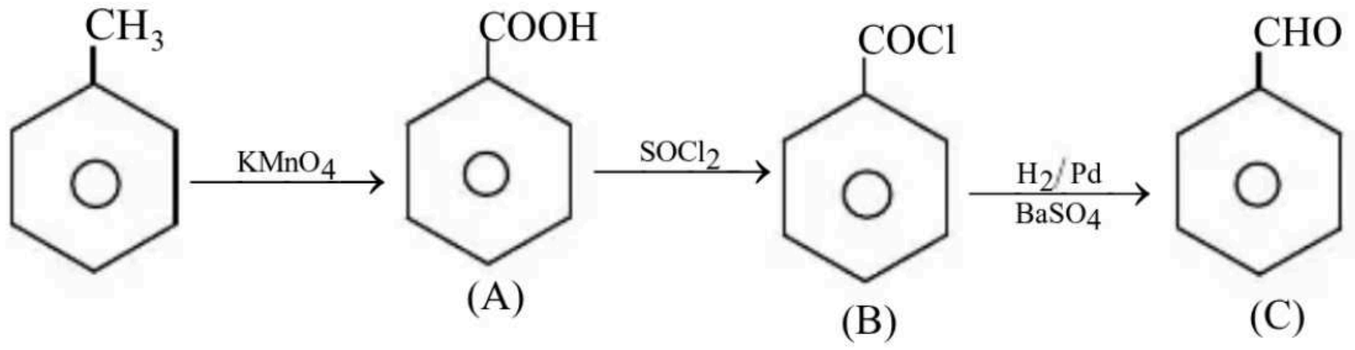
175. $CH_3COOH \xrightarrow{LiAlH_4} \underset{(A)}{CH_3CH_2OH} \xrightarrow{P/I_2} \underset{(B)}{C_2H_5I} \xrightarrow{Alc.KOH} C_2H_4$

176. $CH_3CONH_2 \xrightarrow{LiAlH_4} \underset{\text{Ethanamine}}{CH_3CH_2NH_2}$

177.



178.



179. Glucose has no reaction with Schiff's base, NH_3, $NaHSO_3$

180. Ethanol is used as solvent in making transparent soaps